

Carlos Machicado

Backend & Applied AI Developer | Java · On-Prem LLMs · RAG · Privacy-by-Design
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PROFESSIONAL SUMMARY

Backend Developer with 5+ years of experience and a strong focus on applied Artificial Intelligence. I design and build AI agents and RAG solutions with language models running 100% on-premise (local) and open-source models, on a solid backend foundation in Java 17/21 and Spring. Responsible for architecture and end-to-end development, focused on private, data-sovereign systems —data never leaves to third parties— that are maintainable and secure.

PROFESSIONAL EXPERIENCE

Ricardo Palma University

Backend Developer · Project Management Office

Lima, Peru

Feb 2023 – Present

- Designed and built end-to-end an on-premise conversational RAG agent (Java 21, reactive Spring Boot, Spring AI): hybrid retrieval (vector + Spanish BM25 fused with RRF) and reranking over PostgreSQL + pgvector, mandatory inline citations and layered anti-hallucination/anti-jailbreak guardrails; ~239 classes with a test suite (Testcontainers, k6).
- Served the LLMs locally on a single GPU (vLLM/Ollama, no commercial APIs), sustaining 30 concurrent sessions at p50 7 s, with role-based authorization before retrieval, tamper-evident audit (SHA-256 hash chain) and data-protection compliance (GDPR/LGPD/CCPA). Validated under load testing and entering user testing (pre-pilot).
- As part of the development team, contributed to several modules of the university's academic-management platform (enrollment, teaching and non-teaching load, procurement/requirements planning, dashboards, reporting), on the backend (Java 17/21, Spring) and later the frontend; migrated legacy code (Visual Basic, Java, PHP) to modern Java and Python.
- Contributed to the institutional web-portal revamp, rebuilt the regulatory consumer-complaints module (consolidated, maintainable app) and built automated PDF/Excel reports with Java (Apache POI).

Ganemo

Junior Developer

Lima, Peru (Remote)

Feb 2022 – Nov 2022

- Developed and maintained plugins for the software products across the client portfolio, and automated administrative processes by connecting views and widgets to the database.

Pactomática

Development Trainee

Chile (Remote)

Mar 2021 – Nov 2021

- Automated contracting, HR and contract-management processes ("paperless" initiative), detecting and fixing errors across the automation flows.

TECHNICAL SKILLS

AI & LLMs: AI Agents, RAG, on-premise LLMs (Ollama, llama.cpp, vLLM), Spring AI, LangChain4j, open-source models (Llama, Mistral, Qwen, DeepSeek), vector databases (pgvector, Qdrant, Chroma), NLP, Machine Learning

Backend & APIs: Java 17/21, Spring / Spring Boot, Python, Django, REST API, Apache POI

Databases & Data: SQL Server, Oracle, PostgreSQL, MySQL, Pandas, NumPy

DevOps & Tools: Docker, Docker Compose, CI/CD (GitHub Actions), Git, GitHub, Tomcat

Frontend: React, JavaScript, HTML5, CSS3

Other languages: PHP, Visual Basic

Languages: Spanish (Native) · English (C1)

EDUCATION

M.Sc. in Data Science (completed, degree pending)

Mar 2023 – Jun 2025

Ricardo Palma University

B.Sc. in Mechatronics Engineering

Mar 2014 – Aug 2020

Ricardo Palma University

FEATURED PROJECTS

- NEXUS CORE** — on-premise RAG agent case study: cma208.github.io/nexus-core-study-case
- Deep Learning / ML: customer-churn prediction with convolutional neural networks and tumor classification for breast-cancer detection.